

Beyond One Label per Sentence: Fine-Grained Multi-Label Rhetorical Role Classification in Arabic Court Judgments

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ABSTRACT

Rhetorical role classification (RRC) assigns discourse functions such as facts, legal grounds, reasoning and ruling to the parts of a court judgment, conventionally by labelling each sentence or segment with exactly one role. That convention embeds two assumptions: rhetorical boundaries coincide with the chosen unit, and one unit expresses one role. We test both on a corpus of 200 Arabic judgments (13,038 sentences) from the Dubai Courts, United Arab Emirates, in which two annotators marked variable-length spans with nine roles and an adjudicating legal expert retained multiple roles on one span and nested spans wherever the text warranted them: 7.9% of the 13,574 unique span regions carry more than one role. Projecting the spans of the training and validation judgments onto a sentence grid leaves 20.2% of covered sentences without a unique majority role; a further 7.4% receive a label only by discarding secondary roles, so 27.6% of covered sentences lose adjudicated structure. Fine-tuned Arabic encoders trained on multi-label spans reach a set- F_1 of 0.780–0.781 yet correctly recover only 1.35–1.40 of the 2.02 roles per multi-role span, and forcing a single label costs them 0.12–0.17 set- F_1 on exactly those spans. A derived single-label sentence benchmark remains learnable (macro- F_1 0.72) but only by excluding the structure it cannot express. A blind audit by an independent senior legal expert on 80 stratified items corroborates the adjudicated structure, with only 3 responses disjoint from it. The unit and label-multiplicity assumptions of RRC should be stated and measured, not treated as preprocessing.

1. Introduction

A court judgment is not a flat sequence of statements. It opens with identifying matter, narrates what happened and how the case reached the court, states the charge or claim, records what each side argued, cites the law, reasons about the evidence, and finally decides. Rhetorical role classification (RRC) recovers this functional organisation automatically, and it has become a standard preprocessing step for legal summarisation, retrieval, and outcome analysis [1–4]. Since the earliest CRF systems for Indian judgments, the dominant operationalisation has been to split a judgment into sentences and to assign each sentence exactly one role [3–7].

That formulation is productive. It turns RRC into sequential sentence classification, for which hierarchical encoders, CRF output layers and, more recently, curriculum and graph-based training have delivered steady gains [7–10]. It is also a modelling decision with consequences that are rarely examined. Assigning one label per sentence assumes, first, that the boundaries of rhetorical units coincide with sentence boundaries, and second, that a unit performs one rhetorical function. Neither assumption is obviously true of judicial prose. A court may quote a statutory article inside the sentence that applies it, so that the same words serve as legal ground and as reasoning; a single long sentence may move from the facts to the charge; the operative order may be embedded in a paragraph of reasoning. Where annotation guidelines force one label, these cases are resolved by a rule such as choosing the primary function [8], and the information that a second function was present is lost before any model sees the data. Whether that loss is negligible or substantial is an empirical question, and it cannot be answered from a corpus whose annotation already imposed the one-label constraint.

This paper studies the question on Arabic court judgments whose adjudicated annotation was not forced into a single-label sentence grid. Two annotators independently marked rhetorical spans of arbitrary length with nine roles

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over 200 judgments, and a legal expert acting as adjudicator resolved every disagreement with the explicit option of retaining several roles on one span, or of keeping a shorter span inside a longer one, whenever the text supported both readings. The resulting gold annotation preserves three phenomena that a sentence grid cannot represent: spans that cross sentence boundaries, spans with identical boundaries that carry more than one role, and nested spans with distinct roles. Arabic adds a practical reason to look closely at the unit question. Judgments in the corpus are long, punctuation is sparse and inconsistent, and clause chaining with the conjunction *wa-* produces sentence-like units whose boundaries are themselves a modelling choice [11]; we do not claim that Arabic is unique in this respect, only that it makes the boundary assumption harder to take for granted.

We ask three questions. The first concerns representation: to what extent does the adjudicated rhetorical structure depart from a conventional one-label-per-sentence representation (RQ1)? We answer it by measuring the frequency of multi-role, nested and boundary-crossing structures in the adjudicated annotation and by projecting the annotation onto a deterministic sentence grid and counting what the projection cannot express. The second concerns the original units: given an adjudicated rhetorical span, how accurately can models recover all of its roles, and what is lost when the output is forced to a single label (RQ2)? We treat this as multi-label classification of adjudicated spans (task S1) and compare, on identical model scores, multi-label decoding with forced single-label decoding. The third concerns the conventional formulation: how accurately can the same judgments be classified under single-label sentence-level RRC (RQ3)? For this we build a derived sentence benchmark (task S2) from the same adjudicated spans, keeping only sentences whose projected role is unambiguous, and evaluate lexical, sequential and transformer classifiers on it. S1 and S2 are complementary views of one corpus, not competitors; their metrics answer different questions and are not compared numerically.

The contributions are as follows. (i) An adjudicated rhetorical annotation of 200 Arabic court judgments with nine roles that preserves variable-length, multi-role and nested spans, together with a documented adjudication protocol in which disagreement is never converted automatically into a union of labels. (ii) A quantitative account of how far this structure departs from the sentence-level single-label representation, including a deterministic projection that identifies the sentences a single-label benchmark cannot represent. (iii) A controlled evaluation of multi-label role recovery on the original spans, with a lexical baseline and two Arabic BERT encoders, showing that all systems under-recover label multiplicity and quantifying the cost of one-label decoding on the affected spans. (iv) A conventional single-label sentence benchmark derived from the same gold, on which lexical and CRF baselines are competitive with the transformers, establishing that the conventional formulation is viable for Arabic while making explicit what it excludes. (v) A blind expert validation audit of the structural phenomena, specified and frozen before administration; we report its design, a pilot administration that itself illustrates the single-label habit under study, and the confirmatory senior-expert pass (Section 6.4).

2. Related Work

2.1. Rhetorical role labelling of court decisions

Automatic identification of the rhetorical or functional roles of parts of a judgment originated in extractive summarisation of legal texts, where sentences were classified by rhetorical status to identify summary-worthy content [1, 2, 5]. Saravanan et al. applied conditional random fields (CRFs) over sentences of Indian judgments with seven roles [2, 5], establishing a sequence-labelling formulation that was retained in later work. Savelka and Ashley segmented United States decisions into seven functional and issue-specific parts using CRFs over sentence-level features [12], while Walker et al. compared rule-based scripts with machine learning for sentence roles in veterans' claims decisions [13].

Much of the subsequent benchmark work has focused on English-language Indian judgments. Bhattacharya et al. annotated 50 Supreme Court documents with seven roles and showed that a hierarchical BiLSTM-CRF outperformed feature-based CRFs [3]. Kalamkar et al. scaled the annotation to 354 documents and 40,305 sentences with twelve roles plus a NONE label and introduced transformer baselines [4]. This corpus underlies the LegalEval shared task at SemEval-2023, where rhetorical role labelling was formulated as single-label multiclass classification over sentences and scored by weighted F_1 [6]. Malik et al. released a further 100-document corpus in two legal sub-domains, annotated with 13 fine-grained roles but used a reduced set of seven roles for prediction, and proposed multitask learning with label-shift prediction as an auxiliary objective [8]. Subsequent work has emphasised document context and training strategy. LEGAL-TransformerOverBERT stacks a sentence-level transformer over a legal BERT and reports results on the Indian BUILD benchmark and a confidential Italian corpus with five roles [9]. HiCuLR organises training as a

hierarchical curriculum over documents and roles across four RRC datasets [10], while LegalSeg, the largest resource to date with more than 7,000 documents and 1.4 million sentences in seven roles, benchmarks hierarchical, graph-based, and instruction-tuned models [7]. Outside India, the HKJudge corpus assigns one of 26 roles to each of about 290,000 sentences from Hong Kong criminal judgments; its scheme allows a second tag where an appellate judgment restates lower-court content that keeps a further discourse function, but only 1.97% of sentences carry two tags and its prediction task is formalised as one label per sentence [14].

Across these benchmark formulations, the prediction unit is the sentence and the evaluated target is one role per sentence. The sentence unit itself has also been questioned: Santosh et al. (2023) reformulate rhetorical-role labelling of legal judgments as joint segmentation and classification, using a semi-Markov CRF to identify spans of consecutive sentences that share one role, with gains contingent on the presence of multi-sentence spans [15]. Their spans are contiguous and carry exactly one role, so the formulation moves beyond the sentence but not beyond one label per unit. Agreement is likewise assessed against sentence-level annotations, although the measures differ across corpora. Malik et al. report agreement over sentence labels, whereas Bhattacharya et al. use annotation-difference measures over annotations that were required to cover whole sentences [3, 8]. Role inventories range from five to twenty-six labels but share a recurring core of preamble, facts, issues, party arguments, statute or precedent, reasoning, and ruling, which the Arabic inventory used here also follows.

2.2. Annotation units and label multiplicity

The single-label sentence assumption has not gone unnoticed. Bhattacharya et al. instructed annotators that a label must cover a whole sentence and that a sentence should carry only one label, yet their agreement analysis shows recurring confusion among facts, precedent, lower-court rulings, and judicial reasoning [3]. Malik et al. addressed the issue more directly by allowing annotators to record a secondary and, rarely, a tertiary role when a sentence served more than one function. Secondary roles were assigned in 5–7% of sentences. Their prediction experiments, however, use only the primary label, with the additional labels left for future work [8]. The corpora used by Marino et al. and LegalSeg similarly use one label per sentence, and the HiCuLR curriculum is defined over single-label sentence sequences [7, 9, 10]. The restriction is named explicitly in that line of work: Santosh et al. (2024) list the single-label-per-sentence constraint among the limitations of two papers and propose multi-label reformulation and finer, sub-sentence units as future work [10, 16].

Our treatment of disagreement also connects to a wider re-examination of single gold labels in NLP. Work on human label variation argues that plausible disagreement between annotators is information that aggregation discards, while distinguishing it from annotation error [17], and a survey of learning from disagreement reviews methods that train and evaluate models without assuming a single gold label [18]; perspectivist approaches recommend preserving divergent annotations rather than aggregating them, for instance by majority vote [19]; and shared tasks have scored systems against the distribution of annotator judgments as well as against gold labels [20]. Rhetorical annotation of judgments is a structural instance of the same phenomenon: two experts can both be right about which functions a passage serves. Our protocol is not perspectivist in this sense. It still converges on a single adjudicated layer, and annotator-level variation is reported (Section 3.6) rather than modelled. What it changes is what convergence may produce. Where the adjudicator judged that a passage performs several functions, the gold keeps them as multi-role or nested annotation instead of forcing one label (Section 3.3), and the projection analysis in Section 6 measures what single-label aggregation then discards.

Multifunctionality has therefore been recognised in legal RRC but reduced to a single target in the benchmark formulations reviewed here. Among the resources considered, we found none that jointly preserves variable-length rhetorical spans, admits multiple or nested roles in the gold annotation, measures what is lost when those annotations are reduced to one label per sentence, and evaluates prediction on the original multi-label units. Malik et al. and the HKJudge corpus come closest, because their annotations record a second role on a minority of sentences, but both formalise prediction as one label per sentence [8, 14]. Santosh et al. (2023) share our move from sentences to larger units, but they assign one role per span and evaluate segmentation into contiguous blocks; the co-extensive multi-role and nested structures studied here, and the question of what a single-label projection discards, do not arise in their formulation [15].

Our contribution addresses this representational distinction rather than arguing that sentence-level RRC is unsuitable. S1 retains the adjudicated span structure and its role multiplicity, while S2 derives a conventional single-label sentence representation from the same annotations. Together, S1 and S2 make it possible to measure how much structure is discarded by the reduction and how well the conventional formulation performs after that reduction. The

choice of annotation unit has long been recognised as consequential outside legal NLP. In discourse and argument annotation, segment boundaries form part of the representation design because they determine which distinctions can be expressed and how reliably they can be annotated [21, 22]. Our formulation brings this concern into legal RRC while retaining the sentence-level task as a directly comparable representation.

2.3. Arabic legal NLP and pre-trained encoders

Computational work on Arabic legal text remains comparatively limited, particularly for rhetorical structure. Arabic judicial discourse has nevertheless been studied from a rhetorical perspective. Fakhri analyses the organisation of Arabic fatwas and court judgments through qualitative genre analysis [23]. Recent computational resources also preserve aspects of judicial structure, although at a coarser level. ALARB contains more than 13,000 Saudi commercial cases organised into facts, court reasoning, verdicts, and cited legal provisions [24], while ArabiCCR represents Saudi commercial rulings through case narratives, legal reasoning, and final rulings [25]. These resources provide useful structural information, but their units are coarse or source-defined rather than based on an annotator-defined inventory of fine-grained rhetorical roles.

Other recent work on Arabic judgments addresses related tasks rather than rhetorical role classification. These include a corpus of Saudi Board of Grievances decisions with category labels and abstractive summaries [26], outcome prediction on Saudi commercial court judgments [27], and ruling prediction with large language models [28]. To our knowledge, none provides a fine-grained rhetorical-role benchmark for Arabic court judgments comparable to the RRC resources reviewed above.

Pre-trained encoders for Arabic include AraBERT, trained primarily on Modern Standard Arabic news and Wikipedia [29], and the CAMELBER family, whose MSA variant was evaluated systematically across language variants, training-set sizes, and task types [30]. AraLegal-BERT was trained from scratch on Arabic legal text and reported gains on classification, entity, and keyword tasks [31]. We treat it as related work only because we could not locate an authenticated public checkpoint.

Sentence segmentation is itself a known difficulty for Arabic. Punctuation is inconsistent across genres, and many texts lack reliable boundary markers, which has motivated dedicated Arabic sentence-segmentation resources [11]. This bears directly on the distinction studied here. S1 does not define rhetorical units through automatic sentence boundaries, whereas S2 projects the adjudicated annotations onto a deterministic sentence grid. We therefore treat segmentation as a reproducible component of the S2 representation rather than as a property of the original gold annotation.

3. Corpus and Adjudicated Rhetorical Annotation

3.1. Source judgments

The corpus consists of 200 Arabic court judgments issued by the Dubai Courts (Emirate of Dubai, United Arab Emirates) between 2018 and 2023. They were supplied as extracted plain text with hard line wraps and inconsistent punctuation, which is typical of documents recovered from court file formats. They range from about 1,200 to 116,000 characters (median about 11,500). Split by the deterministic sentence grid of Section 4.2, they contain 13,038 sentences, between 8 and 360 per judgment. Every judgment carries a SHA-256 hash of its text; the hashes are identical across all annotation layers, so no annotator worked on a divergent copy. Two judgments share an identical text, and three others are near-duplicates of one another (word 8-gram Jaccard similarity between 0.48 and 0.77, against a maximum of 0.18 for any other pair). We retain all 200 judgments for the description of the annotation and for agreement analysis, but the exact duplicate pair enters modelling only once (Section 3.5). The example in Figure 2 is synthetic.

3.2. Rhetorical role inventory

The annotation scheme distinguishes nine rhetorical roles (Table 1). Seven of them correspond closely to the roles used in Indian and other common-law corpora: preamble, facts, issue, arguments of each side, cited law, the court's reasoning, and the ruling. Two conventions of the scheme deserve explicit statement because they depart from what a reader of the English-language literature might expect. First, the two argument roles are defined by procedural position rather than by the label a party carries in a given instance: ARGUMENT_PLAINTIFF covers the party pressing the case (plaintiff or prosecution), and by project convention the respondent on appeal, while ARGUMENT_DEFENDANT covers the defending party (defendant or accused) and, by the same convention, the appellant on appeal. Second, DECISION is reserved for the operative merits ruling at first instance, and DECISION_APPEAL covers operative outcomes above first instance or outside the merits, including appeal and cassation outcomes and procedural or prosecutorial orders, even

Table 1

Rhetorical role inventory with guideline-derived definitions and the number of adjudicated annotation rows carrying each role in the full 200-judgment corpus after normalisation (a row is one role assigned to one span; rows with identical boundaries are later collapsed into multi-role regions).

Role	Definition (abridged from the annotation guidelines)	Rows
PREAMBLE	Opening material before the substantive narrative: court and case identification, session and panel, formal opening formulae, introductory recital.	747
FACTS	Neutral narration of the case background and procedural history, without the court's evaluation or a party's request or defence.	3,452
ISSUE	The criminal charge or count, or the precise civil claim or question that the court must decide.	1,445
ARGUMENT_PLAINTIFF	Statements, submissions and requests of the claimant side (plaintiff or prosecution); by convention, the respondent on appeal.	592
ARGUMENT_DEFENDANT	Statements, denials, defences and requests of the defending side (defendant or accused); by convention, the appellant on appeal.	1,039
LAW_REFERENCE	A unit whose function is to cite or quote legal authority: statutes, articles, regulations, decrees-law, or explicitly attributed settled principles.	1,864
ANALYSIS	The court's own evaluative reasoning: weighing evidence, assessing credibility, answering defences, applying law to facts.	4,564
DECISION	The operative part of a first-instance judgment: the merits ruling and what the court orders.	652
DECISION_APPEAL	Operative outcomes above first instance or outside the merits, including appeal or cassation outcomes and procedural or prosecutorial orders.	331
Total rows (200 judgments, after normalisation)		14,686

when such orders occur in a first-instance file. Both conventions are stated in the project's annotation guidelines and were applied consistently by the adjudicator. Named-entity annotations produced in the same project are outside the scope of this study and are not used.

3.3. Independent annotation and adjudication

Annotation proceeded in two stages (Figure 1). In the first stage two annotators, A and B, each annotated all 200 judgments independently in a span-annotation tool. Annotators selected any contiguous stretch of text and assigned it a role; spans were not constrained to sentence or paragraph boundaries, several spans could cover the same text, and one span could be given more than one role. Character offsets into the shared text are the authoritative representation of every annotation, and all statistics in this article are computed from offsets rather than from tokens. Annotator B worked blind to A's annotations.

The two layers disagree substantially, in the way one would expect from unconstrained span annotation. Section 3.6 quantifies this; the short description is that A and B often agree on which words carry a rhetorical function but frequently differ on where a unit begins and ends, on whether a long stretch is one unit or several, and, for a minority of regions, on which role applies. Some disagreements reflect differences of annotation style rather than of legal reading. Annotator A tended to mark a quoted statutory article inside a passage of reasoning as a single span carrying both ANALYSIS and LAW_REFERENCE, while annotator B tended to mark a long ANALYSIS span with a shorter LAW_REFERENCE span nested inside it. Both encodings say that the quotation functions as legal ground within the reasoning; they differ in whether that fact is represented by two labels on one span or by two spans.

All three are legal professionals: A, B and C are native Arabic speakers, lawyers and legal consultants by profession, each holding a law degree and each with over thirty years of professional experience in legal practice. C additionally served as the project's adjudicator as described below.

In the second stage a legal expert, C, acting as adjudicator, resolved the two layers judgment by judgment in a dedicated adjudication tool (SpanArbiter, publicly available at <https://github.com/snkohail/spanarbiter>). C saw both layers with their provenance. Spans on which A and B agreed exactly, in boundaries and role, were accepted automatically. Every other span was surfaced as a conflict, grouped by overlap, and C decided each conflict explicitly: keeping one annotator's span, editing its boundaries, changing or adding a role, deleting a span, deferring the decision, or retaining several roles on one span when C judged that the text genuinely performs several functions. The tool did not merge the two layers automatically: where A and B assigned different roles to overlapping text, the result is whatever C decided, never the union of the two labels by default. No conflict was left deferred in the completed annotation, and

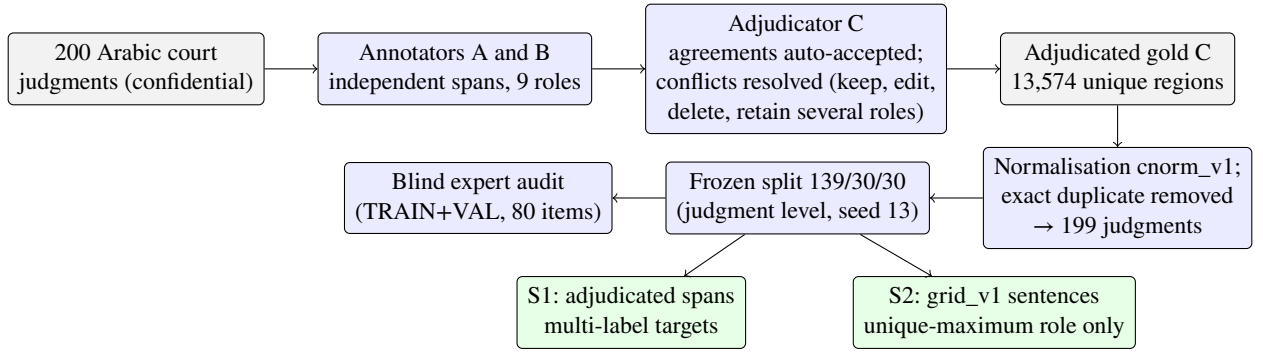


Figure 1: Annotation, adjudication and data-construction pipeline. All 200 judgments are annotated and adjudicated; the exact duplicate is removed and the split is frozen before any model is trained; S1 and S2 are two views of the same adjudicated annotation; the expert audit samples only from the training and validation judgments.

a final validator confirmed that all 200 judgments are complete, that every offset lies inside the text, that every label belongs to the inventory, and that no two rows are exact duplicates. C is therefore an adjudicator with full knowledge of A and B, not a third independent annotator, and we refer to the completed layer as the adjudicated gold annotation rather than as ground truth. An earlier automatic union of A and B, produced during tool development, is not adjudicated and is not used anywhere in this study.

The adjudicated annotation retains three structural phenomena that matter for what follows and that we define precisely here. A *region* is a unique pair of character offsets; rows with identical boundaries are collapsed into one region whose target is the set of their roles. A region is *multi-role* (co-extensive multi-role) when that set has more than one element. Two regions are *nested* when one strictly contains the other; nesting is a relation between distinct boundaries and is independent of how many roles either region carries. Two regions *cross* when they overlap without either containing the other. Multi-role and nested structure are different phenomena and are counted separately throughout. Crossing occurs in 50 judgments and is legal under the guidelines; it plays no further role in the analysis.

3.4. Adjudication outcomes

The merge queued 8,740 span-level conflict units for a single adjudicator. Most conflicts were coverage differences rather than contradictions over shared text: 5,592 units (64.0%) were spans proposed by only one annotator (4,538 B-only, 1,054 A-only); the remainder disagreed over shared text—1,523 role-only (17.4%), 546 segmentation (6.2%), 403 boundary (4.6%), 339 nesting (3.9%), and 337 mixed (3.9%).

The merge produced 14,726 gold rows whose provenance shows that adjudication selected from the annotators' analyses rather than synthesising new ones: 3,966 (26.9%) were identical in both annotation layers, 3,653 (24.8%) and 5,866 (39.8%) originate from A and B respectively, 1,137 (7.7%) were relabelled on an annotator-written boundary, and only 104 (0.7%) rest on a boundary introduced during adjudication; overall 8.4% of rows differ from both source layers, spread across 165 of the 200 documents. No row is a silent union of the two layers. Normalisation removed 40 further rows (29 below the length floor, 11 near-identical), leaving the 14,686 analysed rows.

Adjudication was deliberate: median active review time scaled with conflict difficulty—30 s for single-annotator proposals, 40 s for label-only conflicts, and 70 s for conflicts involving both boundary and label—totalling 99 reviewer-hours of active review (about 41 s per conflict on average). Saved decisions were later revisited for only 4%, 7%, and 11% of these shapes respectively, with revisions concentrating on the structurally hardest conflicts.

Two checks address whether the corpus's multi-role structure reflects the phenomenon rather than an adjudicator encoding preference. First, 52 conflicts pitted the two encodings directly against each other—one annotator analysing a stretch as a single multi-role region, the other as nested regions of different roles over overlapping text: the multi-role structure was retained in 40 cases (22 kept both structures, 18 multi-role alone), the nested structure alone in 1, and neither in 11. Second, of the 1,073 gold multi-role regions, 935 (87.1%) trace to an explicitly adjudicated conflict while 138 (12.9%) passed unopposed, that is, without a competing annotation of the same text from the other annotator, whereas nesting shows the opposite profile—1,543 of 1,832 relations (84.2%) were unopposed. Every multi-role region therefore carries an explicit provenance—deliberately adjudicated or unopposed—and none is a silent merge artefact.

3.5. Normalisation, duplicate handling and the frozen split

Two mechanical artefacts of the annotation tool were repaired by deterministic rules applied in code when the annotation is loaded, so that the adjudicated files themselves are never edited. Rule R1 removes rows shorter than five characters or containing no letter (29 rows, mostly stray clicks on a single word or number). Rule R2 removes the inner copy when two rows with the same role nest with a length difference of at most twenty characters, keeping the outer row (11 rows, arising when A and B marked the same unit with the same role and slightly different edges and C kept both). The 40 removed rows touch 26 judgments and are logged individually with their offsets and reason. We deliberately did not apply a third rule that would have removed a role from a multi-role region when an enclosing region already carries that role; such cases (101 regions) are adjudicator decisions and are kept as made.

The judgment pair with identical text is kept in the corpus description and in the agreement analysis, but only one copy (the lower identifier) enters modelling, so the modelling corpus has 199 unique-text judgments. These were divided once, at judgment level, into 139 training, 30 validation and 30 test judgments with seed 13 using group-aware iterative stratification [32] over eleven document-level indicators: the presence of each of the nine roles, the presence of at least one multi-role region, and a tercile bin of the number of regions per judgment. The three near-duplicate judgments were constrained to the same split. The split was hashed and frozen before any model was trained, and the test judgments were not used for any modelling, preprocessing or threshold decision; all such decisions were made on training and validation data. Table 2 summarises the resulting partition.

3.6. Agreement between the original annotation layers

The A and B layers are the only independent annotations of the corpus, and their agreement is informative about how hard the annotation task is and why adjudication was needed. It is not a ceiling for model performance, because the adjudicated gold is not either layer, and it should not be read as a single number, because unconstrained span annotation can disagree in boundary, in unitisation and in role, and these are measured differently. We therefore report three views. Where A and B produced regions with identical boundaries and a single role each (5,002 regions), they assigned the same role in 73.5% of cases (Cohen's $\kappa = 0.656$ [33]; macro- F_1 between the two label sequences 0.695). Boundary agreement is markedly lower: exact boundary matching between the two layers has $F_1 = 0.474$, and 0.339 when the role must also match; relaxing to any token overlap raises F_1 to 0.628, and to intersection-over-union of at least 0.5 gives 0.569 (0.422 with role). At the level of individual tokens with multi-hot role sets, micro- F_1 is 0.536 and Jaccard similarity 0.366. On the 343 identical-boundary regions to which at least one annotator assigned more than one role, the two role sets are identical in only 2.0% of cases, with a set- F_1 of 0.576; A assigned on average 1.40 roles to these regions and B 1.64. Most conflicts were moreover coverage differences rather than contradictions: 64.0% of queued units were spans proposed by only one annotator (Section 3.4). These figures describe different aspects of the annotation and are not comparable with one another; in particular, the gap between role κ and boundary F_1 does not by itself show that boundaries were harder than roles, since the two quantities have different chance levels and different units. What they establish is that the raw layers could not serve as gold without adjudication, that unitisation and boundary placement account for a large share of the disagreement, and that where both annotators saw more than one function they rarely saw exactly the same set.

4. From Adjudicated Spans to Two Classification Formulations

The adjudicated annotation can be turned into a classification problem in more than one way, and the choice determines what the classifier is asked to do and what the gold standard is able to express. We use two formulations on the same 199 judgments and the same frozen split. The first keeps the adjudicated spans as classification units. The second reconstructs the conventional sentence-level single-label task from the spans. Figure 2 illustrates, on a synthetic example, why the two are not equivalent.

4.1. S1: role recovery on adjudicated spans

In task S1 the unit is the unique adjudicated region and the target is its set of roles, encoded as a nine-dimensional multi-hot vector. A region with roles {ANALYSIS, LAW_REFERENCE} is one example with two positive dimensions. The span boundaries are given; S1 is a classification task, not span detection, and it therefore isolates the question of role recovery from the question of segmentation. Because regions vary from a few words to entire judgments, S1 also exposes the classifier to units that no sentence-level formulation contains, including 319 regions (2.4% of the corpus, 54 in the test set) longer than the 510 sub-word tokens that fit into a standard BERT input. We handle these without

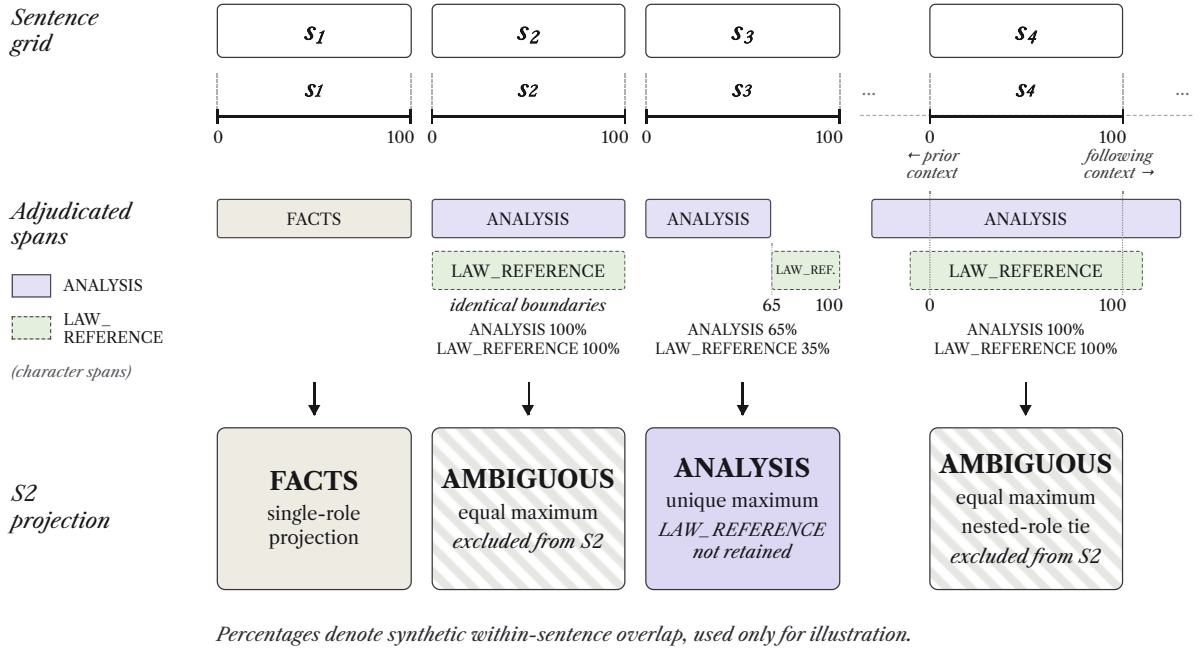


Figure 2: Synthetic illustration (no corpus text) of how adjudicated span structure escapes a one-label-per-sentence representation. Sentence s_1 coincides with a single-role span and projects cleanly (FACTS). A co-extensive multi-role span covers s_2 : ANALYSIS and LAW_REFERENCE tie at identical overlap, no unique majority role exists, and s_2 is excluded as AMBIGUOUS_PROJECTION. The ANALYSIS span continues into s_3 , where a LAW_REFERENCE span begins: the tie breaks (65% against 35% within-sentence overlap), so s_3 receives ANALYSIS and the second role is not retained. Sentence s_4 lies entirely inside a nested pair with different roles: inner and outer spans cover it completely, tie, and s_4 is likewise excluded. Percentages denote synthetic within-sentence character overlap, used only for illustration.

truncation (Section 5.1). The S1 partition has 10,242 training, 1,359 validation and 1,846 test examples (Table 2). Two pre-declared slices of the test set are reported alongside the aggregate: single-role versus multi-role regions, and regions that participate in a nesting relation versus those that do not. The slices are descriptive; they are not matched on any other property and support no causal reading.

4.2. S2: a derived unambiguous single-label sentence benchmark

Task S2 reconstructs the conventional formulation. It requires a sentence grid that does not depend on the annotation, and we use a deterministic, text-only rule that is fully specified in the released code: a blank line is a paragraph break, a single line break inside a paragraph is treated as a soft wrap (the source texts are hard-wrapped extractions), a sentence ends after a full stop, Arabic or Latin question mark, or exclamation mark followed by white space, and a candidate is kept if it contains at least one letter or digit. The grid is hashed and frozen together with the split. It contains 13,038 sentences over all 200 judgments and 12,960 over the 199 modelling judgments, in which the exact duplicate appears once. We prefer this reproducible rule to the unit segmentation produced by the annotation tool, which 89% of adjudicated boundaries happen to coincide with and which would therefore make the projection look far less lossy than an annotation-independent sentence split does. The platform's own units are not a usable alternative grid: they reproduce the source's line wrapping (94% of boundaries newline-adjacent, 39% at a sentence terminator) rather than sentence structure. A more sophisticated segmenter [11] would change the grid but not the logic of the projection; Section 6.1 tests this directly with a neural segmenter.

Each grid sentence is then labelled from the adjudicated regions by greatest character overlap. For every role, the overlap is the number of sentence characters covered by the union of regions carrying that role, so that nested regions with the same role are not counted twice. If exactly one role attains the maximum overlap, the sentence is *eligible* and receives that role. If two or more roles tie for the maximum, the sentence is marked AMBIGUOUS_PROJECTION and

Table 2

Corpus, split and task statistics. Regions are unique adjudicated boundaries after normalisation. S2 counts refer to the frozen grid; only eligible sentences enter S2 training and evaluation.

	Train	Validation	Test	Total
Judgments (modelling corpus)	139	30	30	199
regions per judgment, median (min–max)	40 (4–485)	40 (9–125)	44 (7–184)	
S1: adjudicated regions	10,242	1,359	1,846	13,447
multi-role regions	734 (7.2%)	156 (11.5%)	165 (8.9%)	1,055
regions in a nesting relation	1,273 (12.4%)	328 (24.1%)	269 (14.6%)	1,870
regions > 510 AraBERT tokens	223	42	54	319
S2: grid sentences	9,283	1,764	1,913	12,960
eligible (unique maximum-overlap role)	6,298 (67.8%)	1,261 (71.5%)	1,355 (70.8%)	8,914
AMBIGUOUS_PROJECTION (excluded)	1,534 (16.5%)	377 (21.4%)	322 (16.8%)	2,233
UNLABELED (excluded)	1,451 (15.6%)	126 (7.1%)	236 (12.3%)	1,813

excluded: the annotation gives no principled way to prefer one of the tied roles, and choosing one by a tie-breaking rule would manufacture a label the adjudicator did not assign. A sentence overlapping no adjudicated span at all is UNLABELED and also excluded; we do not introduce a NONE class, because the absence of a rhetorical annotation is not an adjudicator's judgment that the sentence has no role, the span annotation being non-exhaustive by design. For each ambiguous sentence the code records a structural reason available directly from the regions: a single identical-boundary region carries all tied roles (co-extensive multi-role), the tied roles come from regions in a containment relation (nested), or the tie arises from partially overlapping regions (other). These reasons are reported under RQ1.

The result is a *derived unambiguous single-label sentence benchmark*: 6,298 training, 1,261 validation and 1,355 test sentences, each with one of the nine roles. Its labels are projections of adjudicated spans, not an independent sentence-level annotation, and we do not call them sentence gold. Its counts of excluded sentences are themselves a measurement of what the conventional representation cannot carry over from the adjudicated one, and Section 6.1 reports them for the training and validation judgments, on which every methodological decision was taken.

4.3. Blind independent expert validation audit

The structural claims of this paper rest on one adjudicator's decisions. An adjudicator who sees both layers may, in principle, retain two roles or a nested pair because two annotators proposed them rather than because the text warrants them, and the A/B style difference described in Section 3.3 shows how the same legal reading can surface as either structure. To test whether the phenomena are defensible independently of C, we designed a blind expert validation audit and froze it before any expert saw an item. The audit is not a further annotation round, not an inter-annotator agreement study, not a model error analysis, and it does not modify the gold annotation; disagreements it records are reported, not acted upon.

Eighty items were sampled deterministically (seed 13) from the training and validation judgments only, in four strata of twenty: single-role regions not involved in nesting (a control for whether the expert independently recovers ordinary labels); adjudicated multi-role regions; nested pairs whose outer and inner regions have different role sets, preferring pairs in which both regions are single-role so that the two phenomena are not confounded; and AMBIGUOUS_PROJECTION sentences. Caps of two items per judgment per stratum and on label combinations prevent any judgment or role from dominating a stratum. No test judgment and no model output was used in sampling. The expert-facing page shows Arabic passages with the target span or spans highlighted, compact context by default with controls to expand the context or view a very long outer span in full, the nine role names with their guideline definitions, and the questions; it shows no A, B or C label, no provenance and no model information, and items are numbered 1 to 80 in a fixed random order with no stratum shown (the internal response keys, visible only in the exported responses and their on-page preview, carry a stratum code that the page does not explain). For strata one and two the expert selects all roles that genuinely apply, with an UNCERTAIN option and no forced single choice. For nested pairs the expert first judges whether both spans are valid distinct rhetorical units (yes, no, uncertain) and then assigns roles to the outer and inner spans separately. For ambiguous sentences the expert judges whether the sentence can be represented faithfully by exactly one role; if yes, which one; if no, which roles apply.

The pre-declared analysis reports each stratum separately: set- F_1 and exact-set agreement against C for role-set items, the proportion of multi-role items for which the expert independently selected more than one role, the proportion

of nested pairs confirmed as two distinct units, the proportion of ambiguous sentences for which one role was judged insufficient, and UNCERTAIN counts. No agreement statistic pools the four strata, no chance-corrected agreement coefficient is computed over the 80 items, and no corpus prevalence is estimated from this deliberately stratified sample. Section 6.4 reports the audit as administered: a pilot administration to two practitioners, and the confirmatory pass by a senior legal expert.

5. Experimental Setup

5.1. Models

We keep the model set deliberately small. The purpose is not to rank Arabic encoders but to establish, for each formulation, how a strong lexical baseline and two widely used Arabic transformers behave, and to compare them under one statistical protocol.

For S1 the baseline (S1-B0) represents each region by TF-IDF weights over character n-grams within word boundaries and trains nine one-versus-rest logistic regressions, one per role. The two transformer systems fine-tune AraBERTv02 [29] (S1-M1) and CAMeLBERT-MSA [30] (S1-M2) with a nine-unit sigmoid output layer and binary cross-entropy. Both encoders are BERT-base models [34]; we use their public checkpoints with the corresponding fast tokenizers on raw, unsegmented text, so that both encoders see identical input. For AraBERT we deliberately use the v0.2 base checkpoint (bert-base-arabertv02), the variant pre-trained without Farasa pre-segmentation, rather than the Farasa-oriented v2 checkpoint: S1 matches gold spans by character offset, and Farasa segmentation inserts clitic-marker tokens that shift every downstream offset, so an unsegmented checkpoint preserves the gold offset alignment by construction. The choice between the two checkpoints was made on validation data under a fixed rule, by which v0.2 would be the primary AraBERT checkpoint if it matched or exceeded v2 under the identical protocol. On the validation judgments v0.2 exceeded v2 on both tasks for every seed, so v0.2 was fixed as the primary checkpoint. The test-set comparison of the two checkpoints is reported for documentation only and did not determine the selection. On test, AraBERTv02 reaches set-F₁ 0.780 against 0.769 on S1 and macro-F₁ 0.728 against 0.711 on S2 (means over three seeds; on S1 one of the three seed-wise paired intervals excludes zero, on S2 the seed-wise paired differences are +0.036 [0.007, 0.081], +0.007 [-0.015, 0.039] and +0.009 [-0.021, 0.051]), and the v2 numbers appear only in this comparison. Regions longer than 510 sub-word tokens are encoded by an identical procedure for both models: the token sequence is cut into overlapping windows of 510 tokens with a stride of 128 (the last window always reaches the end of the region), each window is encoded as a separate sequence with its own [CLS] and [SEP] tokens, the [CLS] states of all windows are averaged into one region representation, and gradients flow through every window. Regions that fit into one window, 97.6% of the corpus, pass through the same code path with a single window, so the model reduces to the usual [CLS] classifier for them. No region is truncated.

For S2 the systems are a majority-class baseline (S2-B0), the same character n-gram TF-IDF representation with multinomial logistic regression (S2-B1), a linear-chain conditional random field [35] with lexical features (S2-B2), and AraBERTv02 and CAMeLBERT-MSA sentence classifiers with a nine-way softmax and cross-entropy (S2-M1, S2-M2), sharing the encoder, window and optimisation code of S1. The CRF operates on sequences of sentences, and its sequence construction requires care because ambiguous and unlabelled sentences are excluded from S2: a sequence is a maximal run of grid-adjacent eligible sentences of one judgment, and any excluded sentence breaks the sequence, so the model never learns a transition between sentences that were not adjacent in the judgment. This yields 877 training runs with a mean length of 7.2 sentences (164 of them singletons) and 198 test runs. Features are drawn from the sentence itself (word unigrams, first and last words, first-word prefix, length buckets, digit and punctuation flags, initial *wa-*), together with run-boundary flags and the sentence's position decile within the full judgment grid; no text from neighbouring sentences is used, so the CRF's use of context is limited to the learned label transitions.

5.2. Selection, decoding and evaluation

All selection — of hyperparameters, decision thresholds and the encoder checkpoint — was done on the training and validation judgments, and no test judgment was inspected for any selection decision. The lexical models were tuned over a small pre-specified grid (n-gram range {2–4, 3–5}, regularisation $C \in \{0.3, 1, 3, 10\}$, balanced or uniform class weights; for the CRF $c_1, c_2 \in \{0.05, 0.1, 0.5\}$). For the transformers the learning rate was chosen from $\{2 \times 10^{-5}, 3 \times 10^{-5}\}$ with seed 13, up to five epochs with the best validation epoch retained, and the chosen rate was then used for seeds 37 and 73; batches held 16 units or at most 48 windows, with AdamW, 10% linear warm-up, weight

decay 0.01, dropout 0.1 and gradient clipping at 1.0. Every transformer result is reported for the three seeds and as mean and standard deviation.

S1 decoding follows one convention for every system. A single global threshold, selected on validation from the grid 0.05, 0.10, ..., 0.95 by set- F_1 , is applied to the nine probabilities; every role above the threshold is predicted, and if no role passes, the role with the highest probability is predicted, since gold sets are never empty. No per-role thresholds are used. Choosing one constant threshold to optimise a multi-label measure is a standard threshold-calibration strategy [36]; tuning nine role-specific thresholds on 30 validation judgments would multiply the selection space and risk overfitting the rarer roles, so we keep one auditable operating point per seed and return to per-role thresholds in Section 7. The validation split is, if anything, richer in the phenomenon that matters than the test split — 11.5% of validation regions are multi-role, against 8.9% in test (Table 2) — so threshold selection is not starved of multi-role evidence. The primary S1 metric is example-based set- F_1 : for each region, the F_1 between the predicted and gold role sets, averaged over regions. Secondary metrics are macro- F_1 over the nine roles, micro- F_1 over all region-role decisions, the exact-set match rate, per-role F_1 , and the mean number of predicted versus gold roles per region (cardinality), reported overall and on single-role and multi-role regions. The forced single-label comparison re-decodes the same probability matrix by argmax alone and evaluates the result against the same multi-label gold; because inputs, model scores, gold and metric are identical, the only manipulated factor is the one-label restriction. A second control removes the objection that argmax re-decoding handicaps a model trained for multi-label output: we additionally train strictly single-label variants of both transformer systems, identical in encoders, windows and optimisation except for a nine-way softmax output trained with cross-entropy against a soft target that spreads probability uniformly over the gold role set, decoded by argmax at inference (protocol in Appendix B). For S2 the primary metric is macro- F_1 over the nine roles, with micro- F_1 (equal to accuracy in single-label classification) and per-role F_1 as secondary measures.

5.3. Statistical protocol

The judgment is the sampling unit throughout. Confidence intervals are document-clustered bootstrap percentile intervals: the 30 test judgments are resampled with replacement 5,000 times (seed 13), the examples of a judgment drawn k times enter the statistic k times, and the 2.5th and 97.5th percentiles are reported. Comparisons between systems, and between the two decoding rules of one system, use the same resamples for both sides and report the interval of the paired difference; we make no inference from the overlap or non-overlap of two separate intervals. For the transformers, each seed is compared with the baseline separately, and the two encoders are compared seed by seed (13 with 13, and so on). We describe a difference as reliable only when its paired interval excludes zero, and we avoid the word “significant”.

6. Results

6.1. RQ1: how far the adjudicated structure departs from one label per sentence

The adjudicated annotation of the 200 judgments contains 14,686 role assignments after normalisation, which collapse to 13,574 unique regions (Table 3). Three findings bear directly on the sentence-level single-label representation.

First, label multiplicity on a fixed span is a minority but pervasive phenomenon. Of the unique regions, 1,073 (7.9%) carry more than one role, and they occur in 184 of the 200 judgments; a corpus-level single-label design would therefore have had to force a choice in almost every document. Most multi-role regions carry exactly two roles (1,047), eighteen carry three and eight carry more. One combination dominates: ANALYSIS together with LAW_REFERENCE accounts for 564 regions, more than half of the total, which is the pattern in which the court quotes or cites the legal ground inside its own reasoning. The next most frequent combinations, FACTS with ISSUE (138), FACTS with either argument role (48 and 44), and ISSUE with PREAMBLE (41), reflect passages in which the narration of events, the charge and the parties’ submissions are formulated in the same words.

Second, nesting is common and only partly reducible to multiplicity. There are 1,832 strict containment relations between regions, in 120 judgments, and 1,870 regions participate in at least one. Two thirds of these relations (1,210) join regions with different role sets, typically a shorter legal citation, argument or factual statement inside a longer reasoning unit; the remaining 622 join regions of the same role, that is, a long unit that the adjudicator also segmented into parts. A sentence-level representation can express neither.

Third, unit length itself departs from the sentence. The median region is 35 AraBERT tokens, comparable to a sentence, but the 95th percentile is 289 tokens and 319 regions (2.4%) exceed the 510-token input window of a BERT

Table 3

Structural phenomena in the adjudicated annotation (200 judgments) and their consequence for the sentence projection (training and validation judgments of the modelling corpus). Percentages of regions are over 13,574 unique regions; percentages of sentences are over the 11,047 grid sentences of the training and validation judgments.

Phenomenon	Count	Judgments
Unique regions	13,574	200
multi-role (co-extensive) regions	1,073 (7.9%)	184
with 2 / 3 / >3 roles	1,047 / 18 / 8	
most frequent: ANALYSIS+LAW_REFERENCE	564	
FACTS+ISSUE; FACTS+argument roles	138; 48, 44	
Nesting relations (strict containment)	1,832	120
between regions with different role sets	1,210	
between regions with the same role	622	
regions participating in nesting	1,870	
Regions longer than 510 AraBERT tokens	319 (2.4%)	
Grid sentences (train+val)	11,047	169
UNLABELED (no adjudicated span)	1,577 (14.3%)	
covered by at least one span	9,470 (85.7%)	
AMBIGUOUS_PROJECTION	1,911 (17.3%; 20.2% of covered)	149
reason: co-extensive multi-role region	1,165	
reason: nested regions of different roles	643	
reason: partial overlaps	103	
eligible for the single-label benchmark	7,559 (68.4%; 79.8% of covered)	

encoder; the longest region spans 21,432 tokens. Conversely, many regions are shorter than a sentence, which is why the same words frequently belong to two units.

The projection onto the frozen sentence grid measures the cumulative effect. The training and validation judgments together contain 11,047 grid sentences. Of these, 1,577 (14.3%) overlap no adjudicated span at all, most of them headings, signatures and list fragments produced by the extraction, leaving 9,470 sentences covered by at least one span. Among the covered sentences, 1,911 (20.2%, or 17.3% of all grid sentences) have no unique majority role and are marked AMBIGUOUS_PROJECTION. The structural reason is available for each: 1,165 ties arise because one co-extensive multi-role region covers the sentence, 643 because the sentence lies inside nested regions with different roles, and 103 from coincidental partial overlaps. Ambiguity is not concentrated in a few documents: 149 of the 169 training and validation judgments contain at least one ambiguous sentence. The derived single-label benchmark therefore retains 7,559 of the 9,470 covered sentences (79.8%; 68.4% of the grid) in training and validation, and 1,355 of 1,913 grid sentences in test. The retained sentences are not bare pluralities, but the usual winner-share statistic overstates how clean they are. Measured over the union of annotated offsets inside each sentence, the winning role accounts for all annotated character mass in 95.3% of the retained training and validation sentences and for at least 80% of it in 97.0% (median share 1.000). That denominator is generous to the projection: a secondary role nested inside the winning span leaves the union unchanged. Measured against the total role-character mass, counting each role's extent separately, the winning role accounts for everything in 90.8% of retained sentences (at least 80% in 93.3%), and measured against the whole sentence in 74.1% (at least 80% in 95.2%). Counted directly, 699 of the 9,470 covered sentences (7.4%) receive a unique label yet intersect at least one further adjudicated role, whose extent the label then discards; in these sentences the discarded roles cover a median of 35% of the sentence, and at least half of it in 213. Unannotated text is not the explanation: the median covered sentence is fully annotated (mean 0.99; 97.8% annotated to at least 90%). The projection's information loss is therefore larger than the exclusion count shows: 20.2% of covered sentences have no unique majority role and are excluded, a further 7.4% are retained only by discarding secondary roles, and 27.6% of covered sentences thus lose adjudicated rhetorical structure under the single-label projection — in character terms, the benchmark's labels retain 60.5% of the corpus's role-character mass. What is lost is precisely the structure that Table 3 counts: multi-role regions, nested regions of different roles, and the sentences over which such units meet. This answers RQ1 quantitatively: the conventional representation reproduces 72.4% of covered sentences without loss, excludes the 20.2% it cannot express without a decision the adjudicator did not make, and for the remainder keeps the dominant role at the price of the secondary ones.

To test whether this figure depends on the grid, we re-segmented the same 169 judgments with a standard neural segmenter (Stanza, Arabic; grid_v3, frozen and hashed as grid_v1): 23% fewer sentences, median length 180 vs.

Table 4

Sensitivity of the projection to the sentence grid (training and validation judgments, $n = 169$). `grid_v1`: the frozen deterministic rule of Section 4.2. `grid_v3`: Stanza 1.11.0 Arabic tokenizer (PADT model), text-only input, offsets frozen and SHA-256 hashed by the same procedure. Percentages in parentheses are over all grid sentences of the respective grid.

	grid_v1	grid_v3
Grid sentences	11,047	8,515
Median sentence length (characters)	144	180
UNLABELED (no adjudicated span)	1,577 (14.3%)	1,308 (15.4%)
Covered by at least one span	9,470 (85.7%)	7,207 (84.6%)
AMBIGUOUS_PROJECTION	1,911 (17.3%)	1,349 (15.8%)
reason: co-extensive multi-role region	1,165	858
reason: nested regions of different roles	643	412
reason: partial overlaps	103	79
Ambiguous as share of covered	20.2%	18.7%
Eligible as share of covered	79.8%	81.3%
Judgments with ≥ 1 ambiguous sentence	149	139
Judgments whose ambiguity rate moves > 5 points	—	35 (28 down, 7 up)

144 characters (Table 4). The ambiguous share of covered sentences moves only from 20.2% to 18.7%, the eligible share from 79.8% to 81.3% of covered, and the three-way reason distribution is preserved; 35 of 169 judgments change their ambiguity rate by more than five points (28 down, 7 up). The measurement is therefore stable across two independent grids that segment the texts very differently. The S2 benchmark remains on `grid_v1` under the frozen protocol; `grid_v3` is used for this sensitivity analysis only.

6.2. RQ2: role recovery on adjudicated spans

Table 5 reports S1. The lexical baseline is strong: with character 3- to 5-grams, $C = 3$ and a validation-selected threshold of 0.30, it reaches a set- F_1 of 0.746 (95% CI 0.703 to 0.785) on the 1,846 test regions, recovering the exact role set for 63.5% of them. Validation set- F_1 across its sixteen configurations varied only between 0.700 and 0.737, so the result does not depend on a fortunate choice. Both transformers improve on it. AraBERTv02 reaches 0.780 ± 0.004 over three seeds and CAMELBERT-MSA 0.781 ± 0.007 , with corresponding gains in macro- F_1 and exact-set match. The paired comparisons in Table 6 show how reliable these gains are. CAMELBERT-MSA exceeds the baseline by 0.025 to 0.043 set- F_1 with all three paired intervals excluding zero. AraBERTv02 exceeds it by 0.030 to 0.040 set- F_1 (+0.040 [0.016, 0.060], +0.032 [0.014, 0.049] and +0.030 [0.007, 0.050] for seeds 13, 37 and 73), again with all three paired intervals excluding zero. The two encoders are not reliably separated from each other: AraBERTv02 is ahead for seeds 13 and 73 and behind for seed 37, by at most 0.011 set- F_1 , and none of the three same-seed intervals excludes zero. We therefore read the evidence as showing that both transformers improve on the lexical baseline and not as showing a difference between the encoders.

The more informative part of Table 5 concerns label multiplicity. Every system assigns too few roles to the regions that genuinely carry several. On the 165 multi-role test regions the gold mean is 2.02 roles; the baseline predicts 1.38 on average, AraBERTv02 1.46 and CAMELBERT-MSA 1.51, and of these predictions only 1.22, 1.35 and 1.40 roles per region respectively are correct. Set- F_1 on these regions is respectable (0.70 to 0.77, close to the single-role slice for the transformers) because partial credit is given for recovering one of two roles, but the exact-set match rate, which requires the complete set, falls to 0.30 for the baseline and to 0.39 to 0.43 for the transformers, against 0.67 to 0.71 on single-role regions. At the same time all systems over-assign on single-role regions (1.12 to 1.18 predicted roles against 1.00), so the global threshold trades a little precision on the majority for some recall on the minority and still ends well short of the gold cardinality where it matters. Nested regions are harder than non-nested ones for every system, by 0.049 to 0.077 set- F_1 ; this slice is descriptive, and the nested regions differ from the rest in length, role distribution and position, so we do not attribute the gap to nesting as such.

The forced single-label comparison isolates the cost of the one-label assumption at prediction time. Decoding the same probabilities by argmax alone lowers set- F_1 over all test regions by 0.012 to 0.021 depending on the system and seed, with the paired interval excluding zero in every case. The overall cost is small because 91% of test regions have one role and a correct argmax is then a perfect prediction. On the multi-role regions the cost is between 0.10 and 0.17 set- F_1 for every system and seed, and every paired interval excludes zero by a wide margin (baseline: +0.103, [0.068, 0.143]). The loss has a structural floor: the multi-role gold mean is 2.02 roles, and a single-label prediction that

Table 5

S1 results on the 1,846 test regions (30 judgments). Transformer rows give the mean $\pm$ standard deviation over seeds 13, 37 and 73; per-seed set-F₁ is given in brackets. Confidence intervals are document-clustered bootstrap percentile intervals (5,000 replicates); for the transformers the widest per-seed interval bounds are shown. Slices report set-F₁. Cardinality is the mean number of predicted roles per region against a gold mean of 1.09 overall and 2.02 on multi-role regions. The last two rows give the paired set-F₁ difference between multi-label decoding and forced single-label (argmax) decoding of the same probabilities. Nested slices are reconstructed from span offsets within each judgment: a region is nested when it strictly contains, or is strictly contained in, another region of the same judgment.

	S1-B0 TF-IDF + LR	S1-M1 AraBERTv02	S1-M2 CAMeLBERT-MSA
Validation threshold (per seed)	0.30	0.35 / 0.30 / 0.40	0.30 / 0.30 / 0.15
Set-F₁	0.746 [0.703, 0.785]	0.780 $\pm$ 0.004	0.781 $\pm$ 0.007
per seed		0.786 / 0.778 / 0.776	0.783 / 0.788 / 0.771
per-seed CI bounds		[0.721, 0.831]	[0.719, 0.829]
Macro-F ₁	0.720 [0.669, 0.766]	0.749 $\pm$ 0.003	0.741 $\pm$ 0.011
Micro-F ₁	0.738	0.773 $\pm$ 0.004	0.771 $\pm$ 0.011
Exact-set match	0.635 [0.589, 0.676]	0.680 $\pm$ 0.006	0.655 $\pm$ 0.024
Set-F ₁ , single-role regions ($n = 1,681$)	0.750	0.782 $\pm$ 0.004	0.782 $\pm$ 0.009
Set-F ₁ , multi-role regions ($n = 165$)	0.699	0.758 $\pm$ 0.014	0.773 $\pm$ 0.017
Set-F ₁ , nested regions ($n = 269$)	0.680	0.738 $\pm$ 0.007	0.734 $\pm$ 0.018
Set-F ₁ , non-nested regions ($n = 1,577$)	0.757	0.787 $\pm$ 0.006	0.789 $\pm$ 0.008
Exact-set match, multi-role regions	0.297	0.386 $\pm$ 0.034	0.432 $\pm$ 0.052
Predicted cardinality, all regions (gold 1.09)	1.16	1.15	1.21
Predicted cardinality, multi-role regions (gold 2.02)	1.38	1.46	1.51
Multi-label minus forced single-label, all regions	+0.013 [0.004, 0.022]	+0.012 / +0.018 / +0.018	+0.014 / +0.021 / +0.013
Multi-label minus forced single-label, multi-role regions	+0.103 [0.068, 0.143]	+0.126 / +0.149 / +0.121	+0.135 / +0.135 / +0.169

Table 6

Paired document-clustered bootstrap comparisons of test set-F₁ in S1 (difference and 95% interval, identical resamples on both sides). Transformer seeds are compared individually; the two encoders are paired seed by seed.

Comparison	Seed	Difference	95% interval
AraBERTv02 minus TF-IDF baseline	13	+0.040	[0.016, 0.060]
	37	+0.032	[0.014, 0.049]
	73	+0.030	[0.007, 0.050]
CAMeLBERT-MSA minus TF-IDF baseline	13	+0.037	[0.018, 0.052]
	37	+0.043	[0.026, 0.059]
	73	+0.025	[0.005, 0.042]
AraBERTv02 minus CAMeLBERT-MSA	13	+0.003	[-0.018, 0.024]
	37	-0.011	[-0.022, 0.001]
	73	+0.005	[-0.007, 0.015]

recovers one role of a two-role gold set scores exactly 2/3 on that region, so even a perfect single-label decoder caps at 0.667 on these regions — below the 0.70 to 0.77 that multi-label decoding of the same models already achieves there. Since the two decodings use identical inputs, scores and gold, this difference is attributable to the output restriction alone and not to model quality. It also bounds what multi-label decoding currently buys: the systems recover a second role often enough to matter on the affected spans, but, as the cardinality figures show, not often enough to close the gap to the gold.

One objection to the re-decoding control remains: argmax may simply misuse a model whose sigmoid layer was trained for multi-label output. The second control of Section 5 answers it. The softmax variants are single-label by construction — trained with soft-target cross-entropy and decoded by argmax — yet on the multi-role test regions they score 0.11 to 0.18 set-F₁ below multi-label decoding of the corresponding sigmoid model, and all six seed-wise paired intervals exclude zero (Appendix B). The penalty is therefore not an artefact of re-decoding a multi-label model: forcing a single label, whether at decoding time or at training time, is what costs the 0.10 to 0.17. We report this control on set-F₁ only: on a test set that is 91% single-role, the exact-set match of a single-label decoder is mechanically inflated, because one correct label then already constitutes the complete set.

Per-role F₁ (Appendix A) follows the same pattern across systems. PREAMBLE, ANALYSIS, LAW_REFERENCE and DECISION are recovered with F₁ between 0.79 and 0.95; FACTS, ISSUE and ARGUMENT_DEFENDANT lie between 0.65

Table 7

S2 results on the 1,355 eligible test sentences (30 judgments), with per-role F_1 and test support. Transformer rows are means over seeds 13, 37 and 73 (per-seed macro- F_1 in brackets; standard deviations 0.004 for both macro- F_1 rows). Intervals are document-clustered bootstrap percentile intervals.

	Majority	TF-IDF + LR	Lexical CRF	AraBERTv02	CAMELBERT-MSA
Macro-F_1	0.048	0.718	0.722	0.728	0.715
95% interval / per seed	[0.043, 0.053]	[0.640, 0.775]	[0.658, 0.781]	0.734 / 0.725 / 0.726	0.718 / 0.710 / 0.717
Micro-F_1 (= accuracy)	0.278	0.745	0.788	0.765	0.756
PREAMBLE ($n = 262$)	0	0.924	0.964	0.918	0.918
FACTS ($n = 253$)	0	0.648	0.711	0.693	0.686
ISSUE ($n = 118$)	0	0.673	0.759	0.696	0.692
ARGUMENT_PLAINTIFF ($n = 48$)	0	0.413	0.241	0.502	0.442
ARGUMENT_DEFENDANT ($n = 92$)	0	0.670	0.578	0.650	0.584
LAW_REFERENCE ($n = 93$)	0	0.751	0.798	0.768	0.748
ANALYSIS ($n = 376$)	0.434	0.761	0.828	0.797	0.795
DECISION ($n = 74$)	0	0.844	0.853	0.784	0.815
DECISION_APPEAL ($n = 39$)	0	0.773	0.765	0.745	0.752

Table 8

Paired document-clustered bootstrap contrasts of test macro- F_1 in S2 (difference and 95% interval). Transformer seeds are compared individually with each lexical baseline; the encoders are paired seed by seed. Every interval involving a transformer includes zero; the widest per-seed interval is shown.

Contrast	Difference	95% interval
TF-IDF + LR minus majority	+0.669	[0.590, 0.728]
Lexical CRF minus TF-IDF + LR	+0.004	[-0.031, 0.047]
AraBERTv02 minus TF-IDF + LR (seeds 13 / 37 / 73)	+0.016 / +0.007 / +0.008	widest [-0.026, 0.052]
AraBERTv02 minus CRF (seeds 13 / 37 / 73)	+0.012 / +0.003 / +0.004	widest [-0.028, 0.037]
CAMELBERT-MSA minus TF-IDF + LR (seeds 13 / 37 / 73)	+0.000 / -0.008 / -0.001	widest [-0.032, 0.045]
CAMELBERT-MSA minus CRF (seeds 13 / 37 / 73)	-0.004 / -0.012 / -0.005	widest [-0.054, 0.031]
AraBERTv02 minus CAMELBERT-MSA (same seed)	+0.016 / +0.015 / +0.009	all include 0 (widest [-0.015, 0.042])

and 0.78; ARGUMENT_PLAINTIFF, the least frequent role in the training data after DECISION_APPEAL, is the weakest at 0.31 (baseline) to 0.43 (AraBERTv02).

6.3. RQ3: conventional single-label sentence classification

Table 7 reports S2 on the 1,355 eligible test sentences. The conventional formulation is clearly learnable on this corpus. A majority-class baseline that always predicts ANALYSIS reaches a macro- F_1 of 0.048 and accuracy of 0.278. The TF-IDF logistic regression reaches a macro- F_1 of 0.718 (95% CI 0.640 to 0.775) and the lexical CRF 0.722 (0.658 to 0.781) with a higher micro- F_1 of 0.788, reflecting the value of learned label transitions for the frequent roles. The two transformers reach 0.728 ± 0.004 (AraBERTv02) and 0.715 ± 0.004 (CAMELBERT-MSA) macro- F_1 and 0.765 and 0.756 micro- F_1 respectively. Table 8 shows that none of these systems is reliably separated from another: the CRF exceeds the logistic regression by 0.004 with an interval of [-0.031, 0.047], every transformer seed lies within 0.017 macro- F_1 of either lexical baseline with intervals that include zero, and the two encoders are indistinguishable. Only the comparison with the majority baseline is decisive. On this benchmark, therefore, strong lexical and sequential baselines are competitive with fine-tuned Arabic encoders, and we make no claim that the transformers are superior.

The per-role profile in Table 7 is consistent across systems and mirrors S1. Headings, reasoning, legal citations and the operative ruling are recognised well; FACTS and ISSUE are intermediate; the two argument roles are the hardest, and ARGUMENT_PLAINTIFF, with 48 test sentences, is the weakest role for every system, with F_1 between 0.24 and 0.50. The CRF's profile differs from the others in an instructive way: it is the best system on the frequent, sequentially organised roles (PREAMBLE, FACTS, ISSUE, ANALYSIS, LAW_REFERENCE) and the worst on the two argument roles, which are rare and interleaved with facts and reasoning, so that transition modelling helps where the discourse is orderly and hurts where it is not. The test macro- F_1 intervals are wide, roughly ± 0.06 , because 30 judgments with two roles below 50 test sentences leave the macro average exposed to which judgments are resampled. We do not compare the S2 scores with the S1 scores: they are computed on different units with different metrics and answer different questions.

Table 9

Confirmatory expert audit by pre-declared stratum (20 items each). Exact-set counts equality with the adjudicated annotation C; supersets and subsets of C are reported separately and are not exact. Mean set- F_1 is computed against C over the 20 items of the stratum. *Disjoint* counts responses sharing no role with C (for stratum 3, with the outer-span roles). For stratum 4 there is no single gold set: the counts refer to the adjudicated tie, and all 20 responses name roles within or overlapping it.

Stratum	Pre-declared measure	Result	Exact-set vs. C	Mean set- F_1
1. Single-role control	recovers C's role	16 exact, 2 supersets, 2 disjoint	16/20	0.858
2. Multi-role	selects > 1 role	17/20; 13 exact, 3 supersets, 4 subsets	13/20	0.910
3. Nested pair	both units valid	14 yes, 6 uncertain, 0 no	outer 13/20, inner 19/20	outer 0.823, inner 0.983
4. Ambiguous	one role insufficient	15/20 (5 sufficient, 0 uncertain)	9/15 name the tie exactly	—

6.4. Independent expert validation

The audit was administered in two stages. The first was a pilot administration to two practicing lawyers, each with over thirty years of legal practice, who received the instrument exactly as designed. Neither pilot ever selected more than one role on any of the 80 items, although the instrument permits multiple selections throughout, twenty items present adjudicated multi-role regions whose gold sets contain two or more roles, and a further twenty present sentences that the projection itself had to exclude as ambiguous. On the multi-role stratum both pilots returned exactly one role per item. Their choices were nevertheless informed by the structure they declined to mark: on no multi-role item did both pilots choose roles outside the adjudicated set, and on four of the twenty items their two single choices were complementary and together reconstruct the adjudicated pair exactly. The pilot therefore cannot enter the pre-declared agreement analysis — the respondents answered a single-label task the instrument did not pose — and we do not pool it with the confirmatory pass. We report it because it is the paper's thesis observed in the wild: readers habituated to one label per unit produce one label per unit, even when the instrument, the guidelines and the text in front of them all allow more.

The confirmatory pass was administered to a senior legal expert — an associate professor of law and a native Arabic speaker, with over thirty years of professional legal experience — blind to C, to provenance and to stratum. Responses were coded under a rule frozen before counting: a response *confirms* C when its role set equals C's exactly; supersets and subsets of C are reported separately and are not counted as confirmations; for nested pairs, the validity judgement (yes, uncertain, no) and the outer and inner role sets are coded separately; for ambiguous sentences, the sufficiency judgement is coded together with whether the roles named overlap the adjudicated tie; and a response is a *dispute* when its role set is disjoint from C's or when a nested pair is rejected as a single unit. Table 9 reports the coded results by stratum.

On the control stratum the expert reproduced C exactly on 16 of 20 items (mean set- F_1 0.858). The four deviations are two over-selections (two and three roles where C has one) and two disjoint choices (FACTS where C has ANALYSIS; FACTS where C has ARGUMENT_DEFENDANT). On the multi-role stratum the expert independently selected more than one role on 17 of 20 items, reproduced C's set exactly on 13 (one of these responses additionally marked UNCERTAIN), and never returned a set disjoint from C (mean set- F_1 0.910); the remaining deviations are three supersets and four subsets, three of the subsets being a single role drawn from within C's set. On the nested stratum no item was rejected: 14 pairs were confirmed as two valid distinct units and six were marked UNCERTAIN. Inner-span roles match C on 19 of 20 items (mean set- F_1 0.983, the single deviation adding one role), and outer-span roles match on 13 of 20 (0.823); six of the seven outer deviations are over-selections that retain all of C's roles, five of them on items whose validity the expert marked UNCERTAIN, and the seventh is a party-side swap on the outer span, itself marked UNCERTAIN. On the ambiguous stratum, one role was judged insufficient for 15 of 20 sentences; nine of these 15 name exactly the roles between which adjudication had tied, every one of the 15 overlaps the tie, and all five sentences for which the expert judged one role sufficient were assigned a role from within the tie. Three responses of eighty are disjoint from C, two of them in the single-role control stratum: independent disagreement with the adjudication concentrates on ordinary role boundaries, not on the multi-role and nested structure this paper is about. No corpus prevalence is estimated from this sample, and the gold annotation is not changed on the basis of the responses: disagreements the audit records are reported, not acted upon.

6.5. Qualitative inspection of test errors

We inspected the frozen test predictions of the final systems: for S1 the three seeds of S1-M1 (5,538 predictions on the 1,846 test regions), with all patterns checked against S1-M2 and S1-B0; for S2 the three seeds of S2-M1

(4,065 predictions on the 1,355 test sentences), checked against the lexical baselines. Table 10 classifies every S1-M1 prediction as exact, *omission-only* (a strict subset of the gold set), *addition-only* (a strict superset) or *other* (mixed, disjoint or wrong-label). The patterns below hold for every system examined; where systems differ we say so. No judgment passage is reproduced: the analysis is entirely in counts.

On multi-role regions the dominant failure is omission, not commission. Half of all predictions (50.9%) recover a strict subset of the gold set and add nothing, while exactly one prediction in 495 adds a spurious role without missing a gold one. The omitted roles are not random: ARGUMENT_PLAINTIFF is dropped in 83% of its appearances in multi-role gold (75 of 90), ARGUMENT_DEFENDANT in 44%, and FACTS and ISSUE in 40%, while ANALYSIS (18%) and LAW_REFERENCE (26%) — the two roles that dominate multi-role regions — are usually retained. The retained role tends to be the region's discourse-primary function and the dropped role the secondary one. The classifier often nearly selects the role it drops: in 25% of omissions (84 of 331) the dropped role's probability lay within 0.20 below the decision threshold. This is the cardinality gap of Table 5 at the level of individual decisions, and it localises the problem where the discussion of Section 7 places it: not in ranking, since the second role is frequently just below the bar, but in the single global threshold, which is calibrated to keep spurious roles off single-role regions. On those regions the same threshold produces a spurious addition on only 11.0% of predictions. The more than four-fold asymmetry between omission on multi-role regions (50.9%) and addition on single-role regions (11.0%) is the price the current systems pay for protecting the 91% majority, and per-role thresholds or label-dependency models attack exactly this asymmetry.

Where a single-role region is misclassified, the errors cluster in one neighbourhood of the label space. The six largest directed confusions of S1-M1 are ANALYSIS → FACTS (184), ARGUMENT_DEFENDANT → FACTS (78), ARGUMENT_PLAINTIFF → ANALYSIS (64), ISSUE → FACTS (51), FACTS → ANALYSIS (50) and ANALYSIS → ISSUE (46) out of 5,043 single-role predictions: the narrative core of the judgment, where one span reports what a party stated and the next evaluates it. Two confusions one might have expected are essentially absent. LAW_REFERENCE and ANALYSIS are confused in only 33 of 5,538 predictions: quoted legal ground is formulaically marked at span level and the two roles are well separated. Nor do the two project conventions generate confusion: DECISION and DECISION_APEAL are swapped only 19 times, and the cross-party swap between the argument roles is rare and asymmetric (ARGUMENT_PLAINTIFF → ARGUMENT_DEFENDANT 29, the reverse 9).

Nested regions fail differently, and asymmetrically. Inner units behave like ordinary regions: 64.0% exact, with spurious additions on 12.9% of predictions against 9.0% on non-nested regions. Outer units are the problem: only 48.7% exact, with addition-only errors on 31.2% of predictions, about three and a half times the non-nested rate. The added role is almost always the contained unit's own: in 84% of spurious additions on outer regions (81 of 96), the added role is exactly the gold role of the inner unit the region contains. The pattern is not a windowing artefact: the whole-span lexical baseline, which never sees a window, shows the same behaviour (78% of its spurious additions on outer regions are the contained unit's role, against 84–85% for the two encoders), and among the encoders the effect does not grow with the number of windows a region is split into. What predicts it is content rather than architecture: the frequency of inner-role additions increases with the proportion of the outer span occupied by inner text. The nested-slice gap of Table 5 is therefore not evidence that nested units are ill defined — the expert separated every pair without a single rejection, and the inner units themselves are predicted well — but evidence that classifiers respond to rhetorical evidence belonging to a contained unit when predicting the role of its enclosing region; the present analysis does not isolate the mechanism. Document-context encoding, or masking the inner span when classifying the outer one, are natural candidates for isolating it.

S2 shows the same neighbourhood structure at sentence level. The largest off-diagonal cells of S2-M1 over 4,065 predictions are ANALYSIS → FACTS (9.2% of ANALYSIS sentences), ARGUMENT_DEFENDANT → FACTS (18.5%), ISSUE → FACTS (15.0%), FACTS → ANALYSIS (7.4%) and LAW_REFERENCE → ANALYSIS (17.2%), and the same cells dominate the confusion matrices of the TF-IDF and CRF baselines. Two observations follow. First, LAW_REFERENCE, almost never confused at span level, loses a sixth of its sentences to ANALYSIS on the grid: a sentence that quotes a statute inside the court's reasoning carries the marks of both roles, and the single-label grid forces a choice the span formulation does not. Second, ARGUMENT_PLAINTIFF is the weakest role in S2 (recall 0.58) because its sentences disperse into the narrative cluster: of its 61 errors, 30 land on FACTS and 18 on the opposing party's role. These confusions are a property of the label space and the genre rather than of any one system, which is why no system separates from the pack in Table 8.

Across both formulations, then, the residual errors are of two kinds. Boundary confusions inside the narrative cluster (FACTS, ISSUE, ANALYSIS, the argument roles) affect every system alike, and the audit shows that even a senior lawyer disputes them: two of the audit's three outright disagreements with the adjudication sit exactly on this cluster, in the single-role control stratum. Representation-specific failures — omitted secondary roles on multi-role spans,

Table 10

Error taxonomy of S1-M1 (AraBERTv02) on the 1,846 test regions, pooled over seeds 13, 37 and 73 (5,538 predictions; shares in per cent). *Omission-only*: the prediction is a strict subset of the gold set; *addition-only*: a strict superset; *other*: mixed, disjoint or wrong-label errors. Under the argmax fallback a single-role region is never under-predicted, so its failures are wrong labels (18.1%, under *other*; almost all of them a single wrong role) and spurious additions. Nested slices are reconstructed from span offsets within each judgment; 22 regions are both an outer unit of one pair and an inner unit of another.

Slice	<i>n</i>	Exact	Omission-only	Addition-only	Other
Multi-role regions	495	38.6	50.9	0.2	10.3
Single-role regions	5,043	70.9	—	11.0	18.1
Outer unit of a nested pair	189	48.7	2.1	31.2	18.0
Inner unit of a nested pair	684	64.0	1.8	12.9	21.3
Non-nested regions	4,731	69.0	5.0	9.0	16.9

inner-role bleed into outer nested spans, forced choices on mixed sentences — are the failures the formulation itself predicts. The first kind is a property of legal prose; the second is what the multi-label formulation is for.

7. Discussion

Taken together, the three sets of results describe a situation that is more interesting than either a vindication or a refutation of conventional RRC. The sentence-level single-label formulation works well on Arabic court judgments: a character n-gram classifier and a lexical CRF reach macro- F_1 around 0.72 on a derived nine-role benchmark, the transformers do no better, and the per-role profile is the familiar one from Indian and other corpora, with headings and rulings easy and the parties' arguments hard. At the same time, constructing that benchmark required setting aside one covered sentence in five because the adjudicated annotation assigns it two roles with equal claim, or places it inside nested units with different roles. The two observations are not in tension. A representation can be effective for the units it can express and still be unable to express a well-defined part of what human annotators, given the freedom, chose to mark. Our contribution is to have measured that part on one corpus and to have shown that it is neither negligible nor overwhelming: roughly one region in thirteen, fewer than one judgment in ten without any, and one covered sentence in five at the point where the two representations meet.

What is gained by keeping the original spans is the ability to ask a question the sentence benchmark cannot pose: whether a classifier can recover all the functions a unit performs. The answer from S1 is qualified. Set- F_1 in the high 0.70s shows that the roles of adjudicated spans are largely recoverable from their text alone, with no document context; the lexical baseline's 0.746 shows how much of the signal is lexical, and the transformers' modest but reliable gains show what contextual encoding adds. But every system under-recovers multiplicity. Given a span that the adjudicator judged to carry two roles, the systems predict about one and a half, and reproduce the complete set in fewer than half of the cases. The controlled decoding comparison shows that the one-label output restriction is responsible for a loss of 0.10 to 0.17 set- F_1 on exactly these spans, and a control trained natively as a single-label classifier fares no better, losing 0.11 to 0.18 against multi-label decoding of the same encoder: the penalty is representational, not an artefact of how a multi-label model is decoded. The relative magnitudes deserve emphasis. On the multi-role regions, where the representation matters, the choice between the two encoders moves mean set- F_1 by 0.015, while the choice between one label and several moves it by 0.10 to 0.17 — an order of magnitude more. The cardinality figures in turn show that removing the restriction recovers only part of what is lost. The remaining gap is a modelling problem rather than a decoding one: with a single global threshold, the classifier has to raise a second role above the same bar that keeps spurious roles off single-role spans, and the two objectives pull in opposite directions. Per-role thresholds, label-dependency models or calibration might narrow the gap, and we deliberately did not explore them here so that the comparison stays controlled; they are the natural next step for anyone who adopts the multi-label formulation.

Why do the lexical and sequential baselines remain competitive in S2, and close in S1? Part of the answer is the corpus. Judgments from one jurisdiction follow a strongly conventional template with formulaic openings, recurrent connectives that introduce the charge, the defence and the reasoning, and an explicit operative section, so that character n-grams capture much of what distinguishes roles, and label transitions capture the order in which they appear. Part of the answer is size: 139 training judgments and about 6,300 training sentences are small for fine-tuning, and the encoders were used without domain adaptation and with a small, fixed hyperparameter grid, all of which we chose for comparability rather than to maximise their scores. The result does not show that transformers cannot help; it

shows that on this benchmark, under a fair and frozen protocol, they were not demonstrably better than well-tuned classical systems, and that a paper reporting only the transformer results would have overstated what the neural models contribute. The wide document-clustered intervals also matter here: with 30 test judgments and rare roles, differences of a few points of macro- F_1 are simply not resolvable, and the paired intervals say so.

The lower scores on nested regions deserve a similarly careful reading. Nested regions are on average longer, more often multi-role, and more often ANALYSIS or LAW_REFERENCE than the rest, and the slice was not matched on any of these properties. The gap of 0.05 to 0.08 set- F_1 is therefore an observation about a population of regions, not evidence that nesting causes difficulty. The expert audit (Section 6.4) and the error inspection (Section 6.5) together answer the causal question. The independent expert separated every nested pair without a single rejection, so the two units are distinct in the eyes of a legally trained reader. The classifiers fail specifically on the outer unit, and by addition rather than omission: the added role is usually the inner unit's own role, in the window-free lexical baseline almost as often as in the encoders, so the difficulty lies in separating two co-located rhetorical signals, not in the annotation and not in any particular architecture.

Some of what we observe is specific to this annotation scheme and this jurisdiction. The dominance of ANALYSIS with LAW_REFERENCE among multi-role regions is a consequence of a scheme that treats quoted legal ground as a role of its own rather than as part of reasoning, and of a judicial style that quotes statutes inside argument. The two conventions on appeal parties and on orders outside the merits are project decisions that another corpus might make differently. The A/B style difference between two labels on one span and two nested spans is, however, unlikely to be peculiar to this project; wherever annotators are allowed to mark spans freely, the same legal reading can be encoded in either way, and adjudication guidelines need to say which encoding is intended, or to accept both and report them separately as we have. More generally, we think the measurement we report is one that any RRC dataset could report: the proportion of annotated units that a single-label sentence grid can represent without a tie-breaking rule. Datasets whose annotation was collected at sentence level cannot compute it after the fact, which is itself an argument for collecting at least a sample of span-level, multi-label annotation before fixing the unit.

None of this implies that sentence-level RRC should be abandoned. For summarisation, retrieval and navigation, a single dominant role per sentence is often all that a downstream system consumes, the annotation is cheaper and more reliable per unit, and the results here show it is learnable in a new language with modest resources. The implication is narrower: the choice of unit and the single-label constraint are modelling decisions with measurable consequences, and they should be stated and, where the annotation allows, quantified, rather than being treated as preprocessing that happens before the science begins. When the adjudicated annotation of a corpus contains structure the conventional representation cannot express, reporting how much is lost is a small addition to a dataset paper and an honest one.

The audit's pilot administration deserves explicit discussion, because it is the paper's thesis observed in the wild. Two practitioners, shown an instrument that permits multiple roles and items whose adjudicated gold contains multiple roles, returned exactly one role on every item. Their choices were not careless: they stayed within the adjudicated sets, and where the two pilots differed their choices sometimes complemented each other into exactly the adjudicated pair. The structure was visible to them; the single-label habit decided what they recorded. This is precisely the loss the paper quantifies — not that readers cannot see the second function, but that the one-label frame suppresses it at the moment of recording — and it is why the multi-role structure of a corpus annotated under a one-label rule cannot be recovered after the fact. The confirmatory pass shows that the habit is not a ceiling. The senior expert, working through the same instrument blind to the adjudication, selected several roles on 17 of the 20 multi-role items — the behaviour the pilots never produced once — and reproduced the adjudicated set exactly on 13, never returning a set disjoint from C. Not one of the 20 nested pairs was rejected as a single unit, and where the expert hesitated on validity the hesitation surfaced as UNCERTAIN and as over-selection that retains C's roles, not as rejection. Most telling is the ambiguous-projection stratum: for 15 of the 20 sentences that the sentence grid had to exclude, the expert judged that no single role represents the sentence faithfully, nine of the 15 naming exactly the roles between which adjudication had tied; and the five sentences the expert found representable by one role were each assigned a role from within the adjudicated tie. Across all 80 items, only three responses contradict the adjudication outright, and two of those sit in the single-role control stratum: the residual disagreement is the ordinary difficulty of telling FACTS from ANALYSIS or from a party's argument, not disagreement about whether one span can do two jobs or one span can live inside another. The structural phenomena this paper measures are therefore not artefacts of one adjudicator's habits; they are what a legally trained reader reaches for when the instrument does not forbid it.

8. Limitations

The corpus is small by the standards of recent RRC resources and comes from one jurisdiction and one court system. Two hundred judgments suffice to measure the structural phenomena and to train competitive classifiers, but the confidence intervals on test performance are wide, differences of a few points between systems cannot be resolved, and the frequencies we report for multi-role and nested structure describe this corpus and this annotation scheme rather than Arabic judgments in general.

The gold annotation is one adjudicator's resolution of two independent layers. C worked with full knowledge of A and B, and we present C as an adjudicated reference, not as an independent third annotation and not as ground truth. The agreement figures for A and B describe the difficulty of unconstrained span annotation; they are not a ceiling on model performance, and the different agreement statistics we report are not comparable with one another. The blind expert audit is designed to test whether the structural phenomena hold up independently of C, but it is a targeted, stratified validation of 80 items from the training and validation judgments and cannot estimate how prevalent any phenomenon is in the corpus. Its confirmatory weight rests on the senior expert's pass: the two pilot respondents did not engage with the multi-label response mode, so their administrations are reported as an instrument finding rather than as agreement evidence (Section 6.4).

Both task formulations inherit the annotation scheme, and neither is end-to-end. S1 assumes that the adjudicated span boundaries are given and asks only for the roles, so its scores say nothing about how well spans can be detected; a deployed system would need a segmentation step whose errors would compound with the classification errors reported here. S2 is derived from span annotation by a fixed projection rule and a fixed sentence grid; its labels are not an independent sentence-level annotation, its excluded sentences are a property of that rule, and a different grid or tie-breaking policy would produce a different benchmark. We chose the strictest policy, excluding ties rather than resolving them, so that the benchmark contains no label the adjudicator did not assign, at the cost of a smaller and possibly easier test set.

The model comparison is intentionally narrow. Two general-purpose Arabic encoders, one lexical baseline per task and one CRF were trained under a small fixed grid with three seeds; we did not tune per-role thresholds, adapt the encoders to the legal domain, add document context to the sentence classifiers beyond the CRF's label transitions, or evaluate large language models, and the hyperparameter grid was small enough that the transformers' scores should be read as representative rather than as their best achievable. The structural slices (multi-role, nested) are descriptive comparisons of unmatched populations and support no causal claims.

9. Data Availability and Ethical Considerations

The source judgments were provided under a confidentiality agreement and contain personal information about parties, witnesses and officials; they are not released, and no judgment text, in whole or in part, appears in this article, in its figures, or in any accompanying material. The annotation and adjudication tool is available at <https://github.com/snkohail/spanarbiter>. The code for normalisation, split construction, sentence-grid construction, projection, model training, evaluation and the blank bilingual audit instrument, together with the frozen grid and split manifests (judgment identifiers, offsets and hashes only), the aggregate audit and result files, and the frozen test predictions of every reported system, each stored with the gold labels it is scored against, are available at <https://github.com/snkohail/arabic-rrc-adjudicated>. The released predictions and manifests allow every reported model result to be verified without access to the judgments. Annotators and the adjudicator were members of the project team working under the same agreement; the independent expert was given access only to the audit pages, which display passages with highlighted spans and no annotations. No automatic system was used to produce or alter the adjudicated annotation, and the audit responses are recorded but do not modify it.

10. Conclusion

We have examined the representational assumptions behind rhetorical role classification on a corpus of 200 Arabic court judgments whose adjudicated annotation was not constrained to one label per sentence. The annotation shows that about 8% of unique rhetorical regions carry more than one role, that nested rhetorical units occur in 60% of judgments, and that when the annotation is projected onto a sentence grid about one covered sentence in five has no unique role, with one in thirteen more represented only by discarding secondary roles (27.6% of covered sentences lose structure in all); a single-label benchmark built by excluding those sentences retains 68.4% of the grid (79.8% of covered sentences)

in training and validation. Evaluated as multi-label classification on the original spans, a character n-gram baseline and two Arabic BERT encoders recover roles with set- F_1 between 0.75 and 0.78, but all of them predict too few roles on genuinely multi-role spans, and forcing a single output label costs 0.10 to 0.17 set- F_1 on those spans under a controlled comparison. Evaluated as conventional single-label sentence classification on a derived unambiguous benchmark, the same judgments yield macro- F_1 around 0.72 for lexical, CRF and transformer systems alike, with no reliable difference among them. Conventional RRC is thus viable for Arabic and remains a reasonable engineering choice; what the study adds is a way of measuring, and a reason to report, the structure that this choice leaves out. The independent expert validation, specified and frozen in advance, corroborates the structural phenomena independently of the adjudicator: on 80 stratified items a senior expert spontaneously marked multiple roles where the adjudicated annotation does, rejected no nested pair, and judged one role insufficient for three of every four sentences that a single-label projection cannot represent, while the audit's pilot administration illustrates the single-label habit the paper quantifies. The inspection of frozen test errors completes the account: systems fail chiefly by omitting secondary roles on multi-role spans and by assigning the inner unit's role to the outer span of nested pairs — failures of the output restriction and of representations that cannot separate co-located roles, not of the adjudicated annotation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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CRedit authorship contribution statement

Sarah Kohail: Conceptualization, Methodology, Investigation, Data curation, Writing – original draft, Visualization. **Mohamed Emad:** Software, Validation, Formal analysis. **Abdulhamid Yousuf Albalooshi:** Writing – review & editing, Supervision. **Amr Mohamed:** Writing – review & editing, Supervision, Project administration.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used generative AI and AI-assisted tools for language editing and drafting assistance. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

A. Per-role Results for S1

Table A.1 gives per-role F_1 for the three S1 systems.

B. Single-Label Softmax Control

The decoding control of Section 6.2 re-decodes a multi-label model by argmax, and a critic could object that a sigmoid layer trained for multi-label output is not meant to be read as a single-label distribution. The control reported here removes that objection by training classifiers that are single-label by construction. The systems share the encoders (AraBERTv02 and CAMeLBERT-MSA), windowing, optimisation code and hyperparameter-selection rule of S1-M1 and S1-M2, with two changes: the output layer is a nine-way softmax, and the training loss is cross-entropy against a soft target that spreads probability mass uniformly over the gold role set (mass $1/|G|$ on each gold role, so multi-role regions contribute their full set without changing the output arity). Decoding is argmax. Seeds 13, 37 and 73; all selection on the validation judgments. The comparison with the multi-label systems uses the paired document-clustered bootstrap of Section 5.3 on identical resamples.

Table A.1

Per-role F_1 on the S1 test regions under multi-label decoding. Transformer values are means over three seeds. Support is the number of test regions carrying the role.

Role	Support	S1-B0	S1-M1 AraBERTv02	S1-M2 CAMeLBERT-MSA
PREAMBLE	115	0.933	0.939	0.947
FACTS	391	0.650	0.703	0.701
ISSUE	175	0.753	0.780	0.770
ARGUMENT_PLAINTIFF	115	0.311	0.431	0.406
ARGUMENT_DEFENDANT	132	0.683	0.662	0.655
LAW_REFERENCE	267	0.791	0.810	0.814
ANALYSIS	689	0.786	0.837	0.836
DECISION	86	0.830	0.822	0.837
DECISION_APPEAL	44	0.747	0.753	0.706

The comparison metric is example-based set- F_1 only. Exact-set match is uninformative for this control: 91% of test regions carry exactly one role, and for those a single-label decoder needs one correct label to produce the complete set, so its exact-set score is mechanically inflated relative to a multi-label decoder that must also refrain from adding spurious roles.

Table B.1

Single-label softmax control on the S1 test regions: set- F_1 of the soft-target softmax systems against multi-label (sigmoid + threshold) decoding of the corresponding encoder, overall and on the 165 multi-role test regions, per seed, with paired bootstrap intervals for the difference (multi-label minus softmax). Across both encoders and all three seeds, the multi-role difference ranges from 0.11 to 0.18 and every paired interval excludes zero.

Encoder	Seed	Softmax (all)	Softmax (multi-role)	Diff. multi-role	95% interval
AraBERTv02	13	0.774	0.634	+0.130	[+0.090, +0.175]
AraBERTv02	37	0.759	0.630	+0.141	[+0.100, +0.184]
AraBERTv02	73	0.758	0.634	+0.105	[+0.068, +0.151]
CAMeLBERT-MSA	13	0.756	0.612	+0.151	[+0.113, +0.197]
CAMeLBERT-MSA	37	0.759	0.632	+0.127	[+0.094, +0.169]
CAMeLBERT-MSA	73	0.758	0.620	+0.177	[+0.147, +0.214]

The qualitative outcome is stable across encoders and seeds (Table B.1): on multi-role regions the natively single-label systems score 0.11 to 0.18 set- F_1 below multi-label decoding of the corresponding sigmoid model, with all six paired intervals excluding zero; on the full test set the ordering is the same and the differences are smaller (0.01 to 0.03). The one-label penalty is therefore a property of the output restriction, not of the decoding procedure.

C. Normalisation and Grid Rules

Normalisation `cnorm_v1` is applied when the adjudicated files are loaded and never modifies them. R1: a rhetorical row is dropped if its span is shorter than five characters or contains no Unicode letter. R2: if two rows carry the same role, one span contains the other, and their lengths differ by at most twenty characters, the inner row is dropped and the outer kept. R1 runs before R2. Over the 200 judgments R1 removed 29 rows and R2 removed 11, affecting 26 judgments; every removal is logged with judgment identifier, offsets, role and rule.

The sentence grid `grid_v1` is computed from text alone. A paragraph break is a line break followed by optional horizontal white space and another line break; a single line break is treated as white space. Inside a paragraph a sentence ends after one or more of the characters full stop, Arabic question mark, Latin question mark or exclamation mark when followed by white space or the end of the paragraph, so that a terminator glued to the next character (as in dates written with slashes or decimal numbers) does not split. Each candidate is trimmed and kept if it contains at least one letter or digit. The grid over the 199 modelling judgments contains 12,960 sentences and is frozen with a SHA-256 fingerprint over all offsets. The document split (139/30/30, seed 13) is likewise frozen with a manifest hash; both hashes are recorded in the repository and verified before every dataset build.

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